

CSE 232

Computer Networks

Homework-6

Group 15

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Design Implementation

The **print_tcp_option** function is made to handle the Type-Length-Value (TLV) format of TCP options.

Kind values have 2 types:-

- **Kind 0 (End of option list)**- marks the end of all options.
- **Kind 1 (No-Operation)**- A 1-byte placeholder used for padding.

The **while** loop iterates through the options until it finds the end of the buffer or a Kind 0 byte.

- **If (kind==1)**: Just increment the **index i** as it's a 1-byte option.
- **Else if(kind>1)**: First read the length from the next byte.
Then, process the particular option(e.g., kind==5).
Then, advance the index by the entire length of the option to jump to the start of the next one.
- **For kind 5:**
 - No of Ranges (num_ranges) = $(\text{length} - 2) / 8$, where length includes the 1-byte length fields, and after subtracting 2, it gives the byte count of SACK blocks.
 - The size of each block is 8 bytes. So, dividing by 8 gives the number of blocks.
 - $(\text{options} + i + 2 + j * 8)$: Calculates the memory address of the start of a SACK block.
 - $\text{options} + i$: Start of the current SACK option.

- + 2: Skips the Kind and Length bytes.
 - + j * 8: Moves to the start of the j-th 8-byte block
- How much time does it take to transfer a 1 GB file with TCP SACK enabled, and you drop 10% of the packets at the receiver? Use /usr/bin/time scp ... to get the total time.

```
jimmy@BalmerGladen: ~/Desktop/CN/HW6/SACK
~/Desktop/CN/HW6/SACK

jimmy@BalmerGladen | jimmy@BalmerGladen | jimmy@BalmerGl x | jimmy@BalmerGladen

jimmy@BalmerGladen:~/Desktop/CN/HW6/SACK$ /usr/bin/time scp /home/jimmy/Downloads/large.bin ayush@192.168.43.211:/media/ayush/Extra/SACK
ayush@192.168.43.211's password:
large.bin                               57% 585MB 0.0KB/s - stalled -

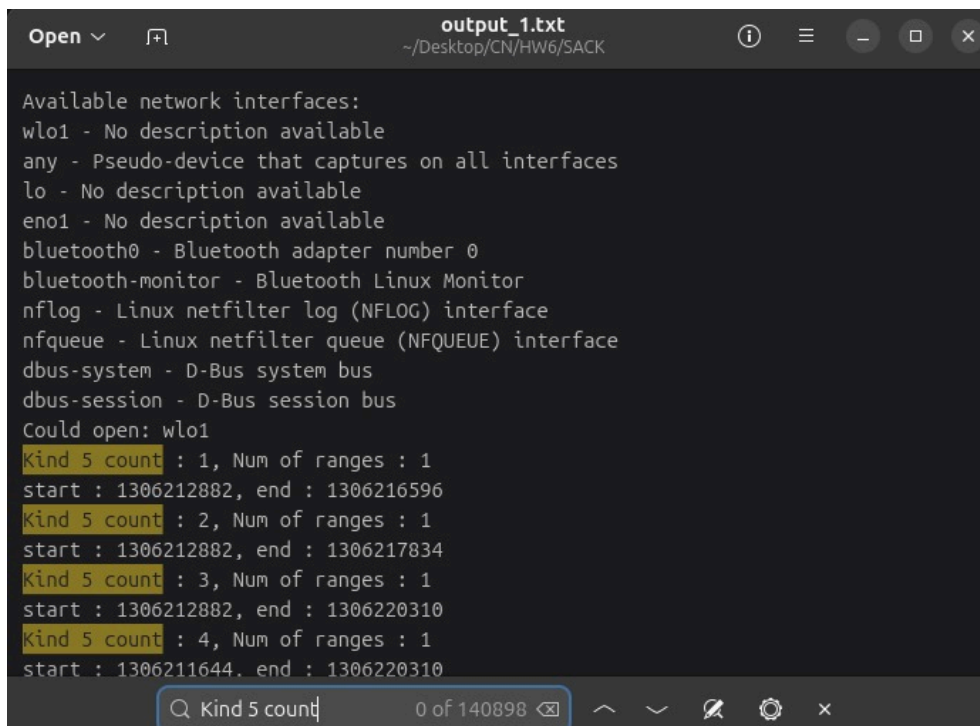
large.bin                               100% 1024MB 281.1KB/s 1:02:10
4.66user 3.52system 1:02:17elapsed 0%CPU (0avgtext+0avgdata 17172maxresident)k
0inputs+0outputs (0major+53185minor)pagefaults 0swaps
jimmy@BalmerGladen:~/Desktop/CN/HW6/SACK$
jimmy@BalmerGladen:~/Desktop/CN/HW6/SACK$
jimmy@BalmerGladen:~/Desktop/CN/HW6/SACK$
jimmy@BalmerGladen:~/Desktop/CN/HW6/SACK$
```

Took **1 hour, 2 minutes, and 17 seconds** to transfer a 1GB file with **TCP SACK enabled**, with an upload speed of **281 kbps**.

- How much time does it take to transfer a 1 GB file with TCP SACK disabled, and you drop 10% of the packets at the receiver?

Took **2 hours, 37 minutes, and 18 seconds** to transfer a 1GB file with **TCP SACK Disabled**, with an average upload speed of **180 KB/S**.

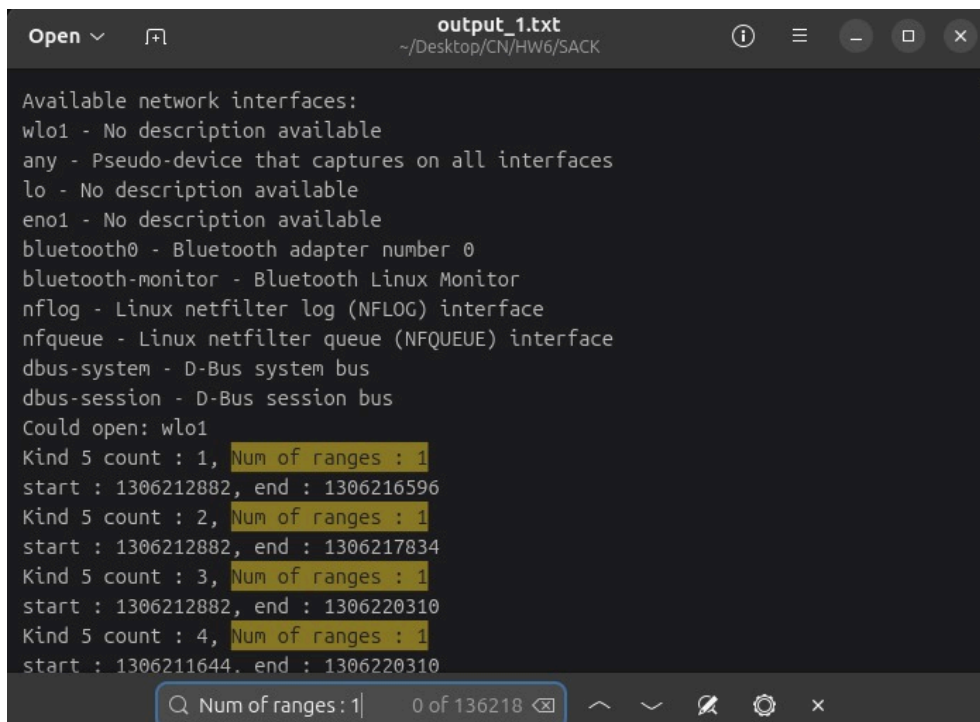
What is the total number of TCP SACK ACKs received during the transfer of the file, with SACK enabled?



The screenshot shows a terminal window titled "output_1.txt" with the path "~/Desktop/CN/HW6/SACK". It lists available network interfaces and shows that "wlo1" was opened. Below this, there are four lines of search results for "Kind 5 count", each showing a count of 1 and a range of start and end values.

```
Available network interfaces:
wlo1 - No description available
any - Pseudo-device that captures on all interfaces
lo - No description available
eno1 - No description available
bluetooth0 - Bluetooth adapter number 0
bluetooth-monitor - Bluetooth Linux Monitor
nflog - Linux netfilter log (NFLOG) interface
nfqueue - Linux netfilter queue (NFQUEUE) interface
dbus-system - D-Bus system bus
dbus-session - D-Bus session bus
Could open: wlo1
Kind 5 count : 1, Num of ranges : 1
start : 1306212882, end : 1306216596
Kind 5 count : 2, Num of ranges : 1
start : 1306212882, end : 1306217834
Kind 5 count : 3, Num of ranges : 1
start : 1306212882, end : 1306220310
Kind 5 count : 4, Num of ranges : 1
start : 1306211644, end : 1306220310
```

- Out of all the TCP SACK received, how many times were the numbers of ranges 1, 2, and 3?



The screenshot shows the same terminal window as before, but with a search for "Num of ranges : 1". The results show four lines, each with a count of 1 and a range of start and end values.

```
Available network interfaces:
wlo1 - No description available
any - Pseudo-device that captures on all interfaces
lo - No description available
eno1 - No description available
bluetooth0 - Bluetooth adapter number 0
bluetooth-monitor - Bluetooth Linux Monitor
nflog - Linux netfilter log (NFLOG) interface
nfqueue - Linux netfilter queue (NFQUEUE) interface
dbus-system - D-Bus system bus
dbus-session - D-Bus session bus
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Kind 5 count : 1, Num of ranges : 1
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Kind 5 count : 3, Num of ranges : 1
start : 1306212882, end : 1306220310
Kind 5 count : 4, Num of ranges : 1
start : 1306211644, end : 1306220310
```

```
Open ▾  output_1.txt
~/Desktop/CN/HW6/SACK

Kind 5 count : 69, Num of ranges : 1
start : 1306713256, end : 1306715732
Kind 5 count : 70, Num of ranges : 1
start : 1306713256, end : 1306716970
Kind 5 count : 71, Num of ranges : 1
start : 1306713256, end : 1306718208
Kind 5 count : 72, Num of ranges : 1
start : 1306724398, end : 1306725636
Kind 5 count : 73, Num of ranges : 1
start : 1306724398, end : 1306726874
Kind 5 count : 74, Num of ranges : 1
start : 1306724398, end : 1306728112
Kind 5 count : 75, Num of ranges : 1
start : 1306724398, end : 1306728112
Kind 5 count : 76, Num of ranges : 1
start : 1306735540, end : 1306736778
Kind 5 count : 77, Num of ranges : 1
start : 1306735540, end : 1306738016
Kind 5 count : 78, Num of ranges : 2
start : 1306739254, end : 1306740492
start : 1306735540, end : 1306738016

Q Num of ranges : 2| 0 of 3826
```

```
Open ▾  output_1.txt
~/Desktop/CN/HW6/SACK

start : 2345207396, end : 2345222252
start : 2345207396, end : 2345222252
Kind 5 count : 1511, Num of ranges : 1
start : 1316629636, end : 1316632112
Kind 5 count : 1512, Num of ranges : 2
start : 2345352242, end : 2345606032
start : 2345207396, end : 2345222252
Kind 5 count : 1513, Num of ranges : 2
start : 2345352242, end : 2345622126
start : 2345207396, end : 2345222252
Kind 5 count : 1514, Num of ranges : 2
start : 2345352242, end : 2345643172
start : 2345207396, end : 2345222252
Kind 5 count : 1515, Num of ranges : 2
start : 2345352242, end : 2345659266
start : 2345207396, end : 2345222252
Kind 5 count : 1516, Num of ranges : 1
start : 1316629636, end : 1316633350
Kind 5 count : 1517, Num of ranges : 3
start : 2345679074, end : 2345701358
start : 2345352242, end : 2345669170

Q Num of ranges : 3| 1 of 3826
```

- **What are the other kinds you have encountered apart from 0, 1, and 5? Briefly describe their purpose.**

- **Kind 2: Maximum Segment Size (MSS)**
 - **Purpose-** It is used only during the three-way handshake at the initiation (on SYN packets).
Each computer uses it to tell the other the **largest amount of data** it is willing to receive in a single TCP segment.
It is done to avoid **IP fragmentation** when a large packet has to be broken into smaller pieces to traverse a network (inefficient).

- **Kind 4: Sack Permitted**
 - **Purpose:** It is also used during the initial handshake.
It's a simple flag where each side announces that they understand and support the use of Selective Acknowledgments.
 - If one side doesn't send it, SACK will not be used for the entire connection. It's the "permission slip" for using SACK.

- **Kind 8: Timestamps**
 - **Purpose:** Contains 2 timestamp values of 4 bytes that serve 2 critical functions:
 1. **Round-Trip Time Measurement (RTTM):** It allows the original sender to calculate the Round-Trip Time (RTT) much more accurately than just timing acknowledgments.
 2. **Protection Against Wrapped Sequence Numbers (PAWS):**
These timestamps help the receiver distinguish between a brand new packet and an old, delayed packet from a previous sequence number cycle, preventing data corruption.